

PHLOP: Physics-Grounded Hierarchical Latent space for Object state and Presentation*

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Abstract—Understanding how objects move, interact, and respond to physical forces remains a key challenge in AI. Existing benchmarks such as CLEVRER, COMPHY, and Physion++ either rely on symbolic labels or lack interpretable grounding for physical dynamics. We introduce PHLOP (*Physics-grounded Hierarchical Latent space for Object state and Presentation*), a diagnostic benchmark that bridges low-level physics and high-level reasoning. PHLOP uses MuJoCo simulations to procedurally generate diverse scenes with realistic object parameters (e.g., mass, friction, elasticity), and augments them with a hierarchical annotations defined by a formal taxonomy, covering different levels of abstraction of physical events (e.g., rolling, sliding, collision). Each video is annotated per-object, per-frame resolution. A question-generation module produces structured queries spanning temporal, causal, and material reasoning. Experiments with state-of-the-art video-language models reveal substantial gaps in physical understanding, mitigated by physics-aware and taxonomy-based prompting. PHLOP enables interpretable, reproducible, and fine-grained evaluation of physical reasoning and supports future research at the intersection of simulation and language.

Index Terms—component, formatting, style, styling, insert

I. INTRODUCTION

Physical reasoning, which involves understanding and predicting how objects interact, move, and respond to forces, is fundamental to both human cognition and machine intelligence (Yi, Wu, Gan, Torralba, Kohli & Tenenbaum 2020, Tung, Ding, Chen, Bear, Gan, Tenenbaum, Yamins, Fan & Smith 2023, Battaglia, Hamrick & Tenenbaum 2013, Piloto, Weinstein, Battaglia & Botvinick 2022a). From an early age, humans intuitively grasp complex physical interactions (Piloto et al. 2022a, Battaglia et al. 2013). For instance, when watching a ball bounce off the floor, people naturally expect it to slow down or bounce back depending on its elasticity, even without explicit calculations (Battaglia et al. 2013, Piloto et al. 2022a). Despite rapid advances in deep learning, AI models continue to struggle with structured and interpretable physical reasoning (Tung et al. 2023, Battaglia et al. 2013, Piloto et al. 2022a, Melnik, Schiewer, Lange, Muresanu, Saeidi, Garg & Ritter 2023, Wang, Li & et al. 2024). Bridging this gap is crucial for applications in robotics, computer vision, and AI, where models must interpret dynamic scenes and make physically plausible predictions (Yi et al. 2020, Tung et al. 2023, Battaglia et al. 2013).

Existing benchmarks, such as CLEVRER (Yi et al. 2020), COMPHY (Chen, Yi, Li, Ding, Torralba, Tenenbaum & Gan 2022), and Physion++ (Tung et al. 2023), have significantly advanced our ability to evaluate models on tasks of physical reasoning; however, each comes with its limitations. CLEVRER relies on symbolic labels for events (e.g., “collision”) that lack explicit physical grounding, making it challenging to analyze fine-grained dynamics (Yi et al. 2020). COMPHY enhances interpretability through compositional physical concepts while simplifying physics by excluding rotational dynamics and approximating friction modeling (Chen et al. 2022). Physion++ integrates a wide range of physical properties and scene diversity, yet it lacks per-frame annotations and event-level semantic labeling, limiting interpretability for temporal and causal analysis (Tung et al. 2023).

Recent work highlights this disconnect, as models struggle to infer latent physical properties from purely visual cues (Hofherr, Sorrenson, Li, Torralba, Schölkopf & Stücker 2023), or match human-level physical reasoning (Piloto & et al. 2024). For instance, while a physics engine may simulate a bouncing ball using precise coefficients, humans describe the same event in qualitative terms: “the ball bounces high because it is very elastic” (Battaglia et al. 2013, Piloto et al. 2022a).

To address these limitations, we introduce **PHLOP** (Physics-Grounded Hierarchical Latent space for Object state and Presentation), a novel benchmark and data set developed from the ground up. PHLOP unifies detailed physical simulation with structured semantic annotations to support hierarchical reasoning over dynamic physical events. Our key contributions include:

- **Physics-grounded Dataset:** PHLOP is a novel benchmark that provides per-object, per-frame data generated from rigid-body simulations. It includes rich physical attributes (e.g., mass, velocity, friction, elasticity) and frame-aligned annotations of physical events grounded in continuous dynamics.
- **Hierarchical physical taxonomy:** We design a structured taxonomy that captures multiple levels of abstraction, from low-level motion and contact dynamics to high-level events, enabling interpretability and flexible supervision.
- **Semantic-Physical Alignment:** PHLOP maps numerical physical behaviors to interpretable labels, bridging the

TABLE I

COMPREHENSIVE COMPARISON OF PHYSICAL REASONING BENCHMARKS. PHLOP UNIQUELY OFFERS DETAILED PER-FRAME ANNOTATIONS DEFINED BY FORMAL TAXONOMY, EXPLICIT FRICTION AND ROTATIONAL DYNAMICS MODELING, AND INTERPRETABLE PHYSICAL EVENT DETECTIONS. BOLD ROWS HIGHLIGHT CAPABILITIES THAT ARE UNIQUELY OR FULLY SUPPORTED BY PHLOP. ✓= FULLY SUPPORTED, -= PARTIALLY SUPPORTED, ✗= NOT SUPPORTED.

Capability / Dataset	PHLOP	Physion++	COMPHY	CLEVRER	ContPhy	AGQA	CATER	PHYRE
<i>General Capabilities</i>								
Question Answering	✓	✓	✓	✓	✓	✓	✓	✗
Diagnostic Annotation	✓	-	-	-	-	✗	-	-
Per-frame Annotations Defined by a Formal Taxonomy	✓	✗	-	✗	✗	✗	✗	✗
Counterfactual Dynamics	✓	-	-	✓	-	✗	✗	✗
Explicit Physical Properties	✓	-	-	✗	-	✗	✗	-
Few-shot Material Reasoning	✓	-	-	✗	✗	✗	✗	✗
<i>Fine-Grained Physical Reasoning</i>								
Real-valued Properties (mass, friction, etc.)	✓	-	-	✗	-	✗	✗	✗
Rotational Dynamics	✓	✗	✗	✗	✗	✗	✗	✗
Friction-induced Events	✓	✗	✗	✗	✗	✗	✗	✗
Elastic vs. Inelastic Collision	✓	-	-	✗	✗	✗	✗	✗
Moving Event Onset Detection	✓	✗	✗	✗	✗	✗	✗	✗
Distribution-based Material Sampling	✓	-	-	✗	✗	✗	✗	✗

gap between low-level physics and high-level reasoning tasks.

- **Model evaluation:** Evaluation of Video-Language models on their physical understanding based on visual and semantic data of object interactions.
- **Reproducible Simulation Framework:** The project includes a GitHub repository that enables the generation of new scenes with controllable physics. PHLOP provides a reproducible method for creating and recreating specific scenes with various object properties and conditions.

II. RELATED WORK

Recent years have seen significant progress in developing datasets that support the study and evaluation of machine understanding of object dynamics, causal structure, and intuitive physics in videos (Yi et al. 2020, Tung et al. 2023). These benchmarks can be categorized as (i) symbolic diagnostic benchmarks, (ii) physics-grounded simulations, and (iii) video-based reasoning task. We briefly review each and position PHLOP within this landscape.

A. Symbolic Diagnostic Benchmarks

The early dataset, such as CLEVR (Johnson, Hariharan, van der Maaten & et al. 2017) and CLEVRER (Yi et al. 2020), primarily focused on symbolic or template-based reasoning. CLEVR (Johnson et al. 2017) established a structured visual question-answering (VQA) system involving object attributes (e.g., “Is there a red cube to the left of the blue sphere?”) using photos of synthetic scenes, but due to its nature, it lacked temporal or physical dynamics. CLEVRER (Yi et al. 2020) was built on this idea and pioneered video-based causal and

counterfactual question answering, around object collisions and motion on the surface. For instance, “What caused the red ball to stop?” or “What would happen if the green object were removed?”. However, physical interactions are abstracted into high-level symbolic labels (e.g., “collide”) without grounding in continuous physical parameters, limiting their utility for fine-grained physical analysis (Yi et al. 2020). Answers to diagnostic questions remain symbolic, lacking annotations that are defined by a formal taxonomy and cover different levels of abstraction, or explicit physical properties to explain causality. IntPhys (Riochet, Castro, Bernard & et al. 2018) introduced violations of intuitive physics, and CRAFT (Wu, Jiang, Sun & et al. 2020) incorporated causal reasoning with forces, but both rely on categorical abstractions rather than physically grounded measures.

B. Physics-Grounded Simulations

To move beyond symbolic abstraction, recent benchmarks have incorporated physics engines to generate realistic dynamics. PHYRE (Bakhtin, van der Maaten, Johnson & et al. 2019) provides a library of 2D physical puzzle scenes governed by gravity, where tasks involve predicting outcomes under known initial conditions. However, PHYRE does not expose object-level physical parameters such as friction or mass.

ComPhy (Chen et al. 2022) expands the scope to more complex 3D multi-object dynamics, supporting compositional physical reasoning (e.g., removing an object and re-simulating). However, it has limited material diversity, representing materials with categorical labels, and simplifies physics by assuming uniform friction and omitting rotational

dynamics and distribution-based sampling of physical parameters.

Physion++ (Tung et al. 2023) further extends these approaches, introducing complex interactions and scene diversity by varying object properties (e.g., size, shape, density) and simulation types (rigid, fluid, deformable). It focuses on high-level prediction tasks (e.g., “Will the object fall off the table?”) rather than detailed physical analysis, and lacks structured semantic grounding of events or per-frame physical state labels (Tung et al. 2023).

Recent approaches like SuperCLEVR (Mao, Gan & Tenenbaum 2024) and PhysGen (Fang & et al. 2024) explore procedural video generation and physics-based prediction, but do not provide per-frame, object-level semantic labels as PHLOP does. InfLevel (Piloto, Weinstein, Battaglia & Botvinick 2022b) measures model surprise under physically implausible conditions (e.g., gravity violation), offering developmental psychology-inspired diagnostics but not per-object tracking.

ContPhy (Zhang, Chen, Zhao & Fan 2024) proposes attribute-based scene analysis (e.g., object softness or size) in physically consistent environments. While this supports high-level reasoning, it does not offer per-frame semantic annotations. Hofherr et al. (Hofherr et al. 2023) explore real-valued parameter estimation using neural implicit functions, and Liu et al. (?) frame physical parameter estimation as an inverse-graphics problem. Both works highlight the need for interpretable, structured metadata like that in PHLOP.

C. Video Reasoning Datasets

Some datasets approach physical reasoning as part of general video understanding. CATER (Girdhar & Ramanan 2019) focuses on object permanence and spatial tracking under occlusion, but uses simple 3D object shuffles without realistic physics. AGQA (Grunde-McLaughlin, Krishna & Agrawala 2021) scales video QA to real-world scenes, asking about action sequences and object interactions, but relies on high-level semantic cues rather than low-level physical simulation. While these datasets advance multi-modal understanding, they are not designed to explore questions requiring physical causality, material properties, or dynamical system behavior.

More recent efforts such as Helping Hands (Cheng, Das & Hauptmann 2023), Learning Object States (Sun & et al. 2023), and disentangled SDE dynamics (Lee, Li, Gan & Tenenbaum 2023) contribute to object-centric video reasoning, highlighting temporal dependencies and state changes in visual data. While these datasets advance multimodal understanding, they are not designed to explore questions requiring physical causality, material properties, or dynamical system behavior—domains where PHLOP aims to bridge the gap.

D. PHLOP in Context

Recent work leverages procedural priors for scene generation and video prediction (Takenaka, Ma, Chang, Santoro & Mirowski 2023), suggesting that control over simulation parameters can guide model learning. The design of PHLOP

is also inspired by theoretical foundations: simulation-based cognition (Battaglia et al. 2013), intuitive physics learning (Piloto et al. 2022a), and physics-guided machine learning (Karniadakis, Wang & Pang 2023). Reference material properties are grounded in established engineering and physics resources (Ashby & Jones 2018, Callister 2020, Halliday, Resnick & Walker 2013), ensuring physical realism in the dataset. PHLOP builds on these advances while addressing critical limitations in existing physical reasoning datasets. Table I outlines how the dataset uniquely bridges low-level dynamics and high-level semantics.

PHLOP uniquely enables:

- **Per-frame semantic labeling** of object states and transitions—absent in all prior benchmarks.
- **Rotational and frictional event detection**, such as rolling, slipping, or friction-induced halts.
- **Elastic vs. inelastic collision analysis** using velocity changes and energy loss metrics.
- **Few-shot material reasoning** via continuous distribution-based sampling across density, elasticity, and friction.
- **Moving event onset detection**, allowing frame-accurate identification of motion transitions.

These capabilities enable novel types of questions such as:

- “Which object experienced the greatest kinetic energy loss?”
- “When did the red ball stop rolling?”

Such queries require the unified physical and semantic grounding that prior datasets lack. By combining continuous dynamics with interpretable labels, PHLOP offers a scalable, diagnostic benchmark for evaluating grounded physical reasoning in vision-language models.

III. METHODOLOGY

In this section, we describe the creation of PHLOP, which involves two core components: (1) the dataset creation process for generating physics-accurate videos with ground truth annotations, and (2) the benchmark evaluation framework for assessing video-language models on physical reasoning tasks.

A. Dataset Creation

1) *Video Generation Process*: PHLOP contains 4,000 generated videos, each depicting a simulated physical scene involving multiple rigid objects with varying properties and interactions. While this initial release is smaller than large-scale datasets such as Physion++ or AGQA, it was constrained by available computational resources. Importantly, the generation pipeline is fully automated and openly released, making it straightforward to scale PHLOP to hundreds of thousands of videos in future work.

Scenes are created using the MuJoCo physics engine (Todorov, Erez & Tassa 2012). Each scene involves one or more physical events such as collisions, sliding, rolling, or frictional deceleration. The simulation captures these interactions at a high resolution of 1024×768 pixels at 25 FPS, capturing

dense temporal sequences of interactions. This object-centric logging bridges physical reasoning and visual understanding by aligning data with physical ground truth.

The dataset is generated using a controlled randomization method over object attributes (shape, size, color), physical properties (e.g., friction and restitution), and initial conditions (e.g., velocity vectors, spatial configurations).

2) *Semantic Annotation and Taxonomy*: Physical events within each video are annotated automatically at the frame and object level using PHLOP’s hierarchical taxonomy. This taxonomy categorizes interactions across several abstraction levels:

- **Level 1**: Broad physical category (e.g., Kinematic Events)
- **Level 2**: Event type (e.g., Rotational Motion)
- **Level 3**: Fine-grained label (e.g., Rolling with Slipping)

Table II defines this taxonomy. It allows PHLOP to support taxonomy-based prompting and structured question generation.

Annotations are derived from simulation logs using rules over motion thresholds, velocity changes, and force-based triggers. For example, an object labeled as “Friction Stop” must meet criteria for velocity approaching zero under non-collisional deceleration.

3) *Parameter Sampling*: To ensure both visual and physical diversity, PHLOP samples a range of object-level and global scene parameters. Tables III and IV summarize the full range of material and scene properties used.

TABLE III
MATERIAL PHYSICAL PROPERTIES WITH REAL-WORLD REFERENCE
VALUES (ASHBY & JONES 2018, CALLISTER 2020).

Material	Elasticity (–)	Density (kg/m ³)	Friction (–)	Examples
Metal	0.85–0.92	2700–19320	0.28–0.32	Steel, Aluminum, Copper
Wood	0.35–0.45	500–700	0.45–0.55	Softwood, Hardwood
Rubber	0.90–0.98	900–1200	0.90–1.10	Natural, Synthetic
Glass	0.55–0.65	2400–2600	0.18–0.22	Soda-lime, Borosilicate
Plastic	0.65–0.75	1000–1800	0.35–0.45	LDPE, PP, PC

Material Physical Properties: Table III presents material-specific property ranges with representative real-world context. Each object in the simulation is assigned properties that determine its physical behavior and visual appearance. Sampling was performed from the material-specific distributions that can be found in Table III. Each object is assigned:

- **Shape**: Chosen randomly from `sphere`, `cube`, `cylinder`, or `block`, each with different moments of inertia and contact surfaces.
- **Material Type**: Samples randomly from a discrete set (e.g., `wood`, `metal`, `rubber`, `glass`, `plastic`), each associated with different parameter distributions.
- **Physical Properties**: Material-dependent distributions for density, elasticity, and friction (Table III).

- **Visual Properties**: Material-specific transparency, specular, shininess, and random color.
- **Dimensions**: Ball/cylinder radius 0.05–0.1m, cube side 0.05–0.1m, block 0.01–0.1m × 0.05–0.1m × 0.05–0.1m.
- **Initial Position & Orientation**: Randomized in 3D space to create spatial variability across simulations.
- **Mass**: Calculated from volume and sampled density, minimum 1e-6 kg.
- **Initial Velocities**: Linear velocities are sampled in meters per second (m/s) and angular velocities in radians per second (rad/s). They uniformly sampled in 3D space $[-1, 1] \times [-1, 1] \times [-1, 1]$, (x, y, z) .

Global Scene Parameters: Table IV provides comprehensive specifications for environmental factors:

- **Lighting**: *Position*, *color*, *reflectivity*, and *angular spread* control the directionality and softness of shadows and highlights.
- **Camera**: *Azimuth* and *elevation* define the viewing angle; *distance* and *look-at point* control framing and focus.
- **Floor Plane**: *Friction coefficient*, *color* (based on friction), *shininess*, and *specular reflectance* are sampled to vary how objects interact with surfaces visually and physically.

All parameters—both global and per-object—are stored alongside the simulation for full reproducibility and analysis. These attributes drive the emergence of behaviors such as gliding, bouncing, tumbling, and frictional halts, allowing PHLOP to capture a comprehensive range of Newtonian phenomena.

The figures 1 and 2 illustrate different scenes from the dataset, where the number of objects and environmental conditions vary. Notably, the camera position and lighting conditions differ between the two scenes. The taxonomy annotations also reflect different physical interactions. Figure 1 shows a more complex scenario involving an inelastic collision, where a moving object interacts with a stationary one, causing it to move. In contrast, Figure 2 presents a simpler case, where two objects slide and eventually stop due to friction.

4) *Verification and Quality Assurance*: To ensure visual-semantic alignment and correctness of the annotations, each simulation produces three video modalities: RGB video, segmentation video (with object-specific masks), and annotated video (with bounding boxes, physical attributes, and semantic event labels). These were used to manually verify that physical events, object identities, and interaction dynamics were captured correctly across time.

While most annotations are generated algorithmically from simulation logs using rule-based logic, approximately 5% of the generated dataset (~200 videos) was manually reviewed to validate consistency. For example, rolling events were cross-checked against angular and linear velocity curves, or stationary objects were checked whether linear velocity is approximately zero within a threshold. Future work may explore using learned classifiers to automate this verification at scale.

TABLE II
HIERARCHICAL TAXONOMY OF PHYSICAL EVENTS IN PHLOP. EVENTS ARE ORGANIZED INTO GENERAL CATEGORIES (LEVEL 1), SUBCATEGORIES (LEVEL 2), AND SPECIFIC EVENT LABELS (LEVEL 3).

Level 1 (General Category)	Level 2 (Event Type)	Level 3 (Specific Event Labels)
Kinematic Events	Linear Motion	Constant Velocity Decelerating Accelerating Stationary
	Rotational Motion	Pure Rotation Rolling Motion Rolling Motion with Slipping
Interaction Events	Collisions	Elastic Collision Inelastic Collision
State Transitions	Motion Change	Moving to Stopping Stationary to Moving
Environmental Interactions	Friction-Induced Events	Friction Stop (Stopped Due to Friction) Sliding with Friction

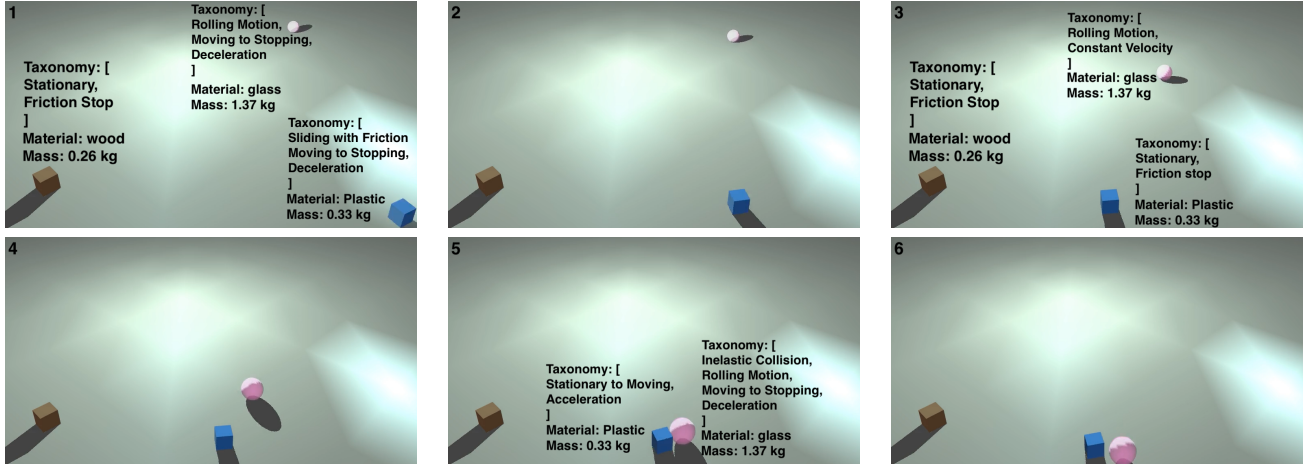


Fig. 1. Sequence of 6 frames showing dynamic object interactions and taxonomy-based annotations in the PHLOP simulation. Each frame captures per-object motion, material properties, and physical events.

B. Benchmark Methodology

1) *Automatic Question Generation*: PHLOP includes a built-in module for automatic question generation based on the annotated physical interactions and object properties. This module leverages simulation logs (mass, velocity, position, etc.) and per-frame taxonomy labels to create grounded question-answer pairs. Questions are designed to test a spectrum of reasoning skills, including:

- **Temporal Reasoning**: e.g., “At what time does the red ball become stationary?”
- **Material Property Inference**: e.g., “Which object had the highest friction coefficient?”
- **Interaction and Outcome**: e.g., “Which object was involved in the most collisions?”
- **Aggregate Events**: e.g., “How many objects rolled during the video?”
- **Energy & Loss**: e.g., “What percentage of system kinetic energy was lost in a collision?”

Unlike symbolic QA in CLEVRER (Yi et al. 2020), PHLOP’s questions are grounded in both semantic labels and physical data. This allows richer evaluation of model understanding across symbolic, causal, and physical domains. The questions and the respective correct answers are unambiguous because of their grounding in the simulation data, enabling evaluation of both the language and the compositional physical understanding.

2) *Prompt Construction and Types*: To explore how structured information impacts physical reasoning, we designed four prompting conditions for model evaluation. Each model receives 32 uniformly sampled frames, an instruction on how to answer, and a question with multiple-choice options.

- 1) **Video Only**: No metadata is included.
- 2) **Physics Only**: The prompt includes object-specific physical parameters, such as:

“1. darkcyan sphere: mass = 0.8, friction = 0.92”

- 3) **Taxonomy Only**: The prompt includes semantic event annotations for each object, based on the formal taxon-

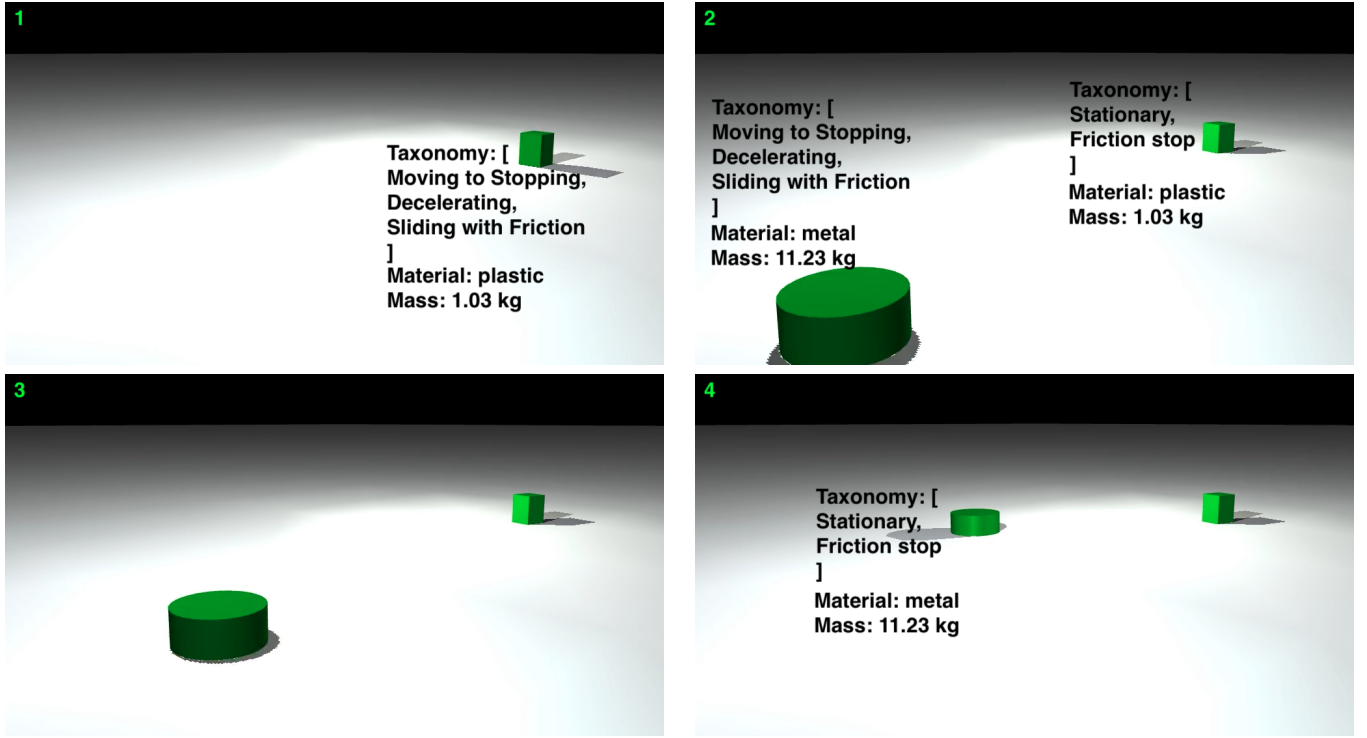


Fig. 2. Sequence of 4 frames showing dynamic object interactions and taxonomy-based annotations in the PHLOP simulation.

TABLE IV
GLOBAL SCENE PARAMETER SPECIFICATIONS

Component	Parameter Range	Description
Camera	Lookat: $[-0.5, 0.5] \times [-0.5, 0.5] \times [-0.5, 0.5]$, Distance: 1.5–3.5 m	Camera position in 3D space
	Azimuth: $[-90^\circ, -30^\circ]$, Elevation: $[-30^\circ, 0^\circ]$	Viewing angle constraints
Lighting	Position: $[-1, 1] \times [-1, 1] \times [0.1, 0.7]$	Lighting position in 3D space
	Diffuse: 0.7–1.0, Specular: 0.5–1.0	Light intensities
Floor	Cutoff: 50° – 180°	Angular spread
	Friction: Static $[0.0, 1.0]$, Dynamic $\leq$ Static	Three-component friction
	Specular: 0.2–0.8, Shininess: 0.2–0.8	Visual settings
	RGBA: brightness	Color mapping

omy (Table II), such as:

“1. darkcyan sphere: Rolling Motion, Friction Stop”

4) **Combined:** Both physical properties and event annotations are included.

These prompts are structured using a dynamic generation script that formats each input with:

- **Object properties:** mass, friction, shape, color
- **Per-frame behaviors:** taxonomy-based semantic labels

• Question classification and answer formatting guidelines

Critically, prompts provide contextual cues but not the correct answer. For example, a label like “Friction Stop” suggests a stopping event, but not when or which object stopped first—this still requires reasoning.

Including taxonomy labels can serve as a partial hint, potentially boosting performance and acting as an upper bound on model capabilities with semantic guidance. To mitigate trivial retrieval, prompts include labels for all objects—even those irrelevant to the current question—introducing distractors. This tests the model’s ability to focus on the relevant information and reason compositionally rather than memorizing facts.

This format mirrors human reasoning: combining perceptual evidence (video) with physical understanding (mass, friction) and semantic interpretation (event labels). The model must synthesize this multimodal input to answer correctly, rather than retrieving answers from explicit labels. This setup investigates whether models can use low-level physics, high-level semantics, or their combination for better QA performance, based on the findings of Physion++ (Tung et al. 2023) and ContPhy (Zhang et al. 2024).

3) *Evaluation Protocol:* We evaluate three state-of-the-art open-source video-language models on PHLOP:

- **InternVideo2.5-Chat-8B** (Wang, Li, Yan, He, Yu, Zeng, Wang, Ma, Huang, Gao, Dou, Chen, Wang, Qiao, Wang & Wang 2025): A large-scale, chat-augmented multi-modal model designed for long-range temporal reasoning in videos. It incorporates rich contextual modeling and

supports complex dialogue-style interaction, making it suitable for temporally extended physical tasks.

- **InternVL3-2B** (Zhu et al. 2025): A lightweight, vision-language transformer optimized for efficient inference. Despite its smaller size, it demonstrates competitive performance on several vision-language benchmarks and offers a strong baseline for zero-shot physical QA under resource-constrained settings.
- **VideoLLaMA3-7B** (Zhang, Li, Cheng, Hu, Yuan, Chen, Leng, Jiang, Zhang, Li, Jin, Zhang, Wang, Bing & Zhao 2025): An advanced, multi-modal LLaMA variant incorporating video encoding modules and extensive pre-training. It integrates spatial and temporal understanding and has shown strong results on general video question-answering tasks, making it a high-capacity candidate for assessing grounded physical reasoning.

All models are evaluated without fine-tuning due to computational constraints. Although the experiments were run on an NVIDIA A100 GPU with 60GB of memory, the resource limits prevented full-scale training and fine-tuning. Thus, we restricted the benchmark to zero-shot evaluation on open-source video-language models that could operate within available hardware limits. Each model receives identical input: the video, question, and optionally physical properties or taxonomy labels or both, according to the prompting condition. This design follows prior work such as CLEVRER (Yi et al. 2020), which used programmatic question-answer templates, and COMPHY (Chen et al. 2022), which measured compositional reasoning via counterfactuals.

IV. RESULTS

A. Overall Model Performance

Results in the Table V show consistent performance gains from structured metadata, particularly when physical and semantic cues are combined. VideoLLaMA3-7B notably achieves the highest accuracy, reflecting its capacity to integrate multimodal information for physical reasoning.

B. Statistical Validation

A two-way ANOVA was conducted with model architecture and prompting condition as independent variables, and Q&A accuracy as the dependent variable. The analysis revealed statistically significant main effects for both factors ($p < 0.001$), as well as a significant interaction effect ($p < 0.001$). These results confirm that model performance is influenced not only by architecture and the availability of structured context (i.e., physical parameters and event taxonomy), but also by how different models benefit from these forms of metadata.

Notably, the interaction effect suggests that certain models are better able to leverage structured information for physical reasoning than others. These findings underscore the diagnostic value of PHLOP and motivate the more detailed breakdown by model, prompting condition, and question type presented in the following sections.

Post-hoc Analysis: We conducted a Tukey HSD post-hoc test to identify which specific model-condition pairs differed

significantly in accuracy. Across the 12 combinations (3 models $\times$ 4 prompting conditions), most pairwise comparisons were statistically significant ($p < 0.001$ after adjustment). The Combined(taxonomy and physics) condition consistently outperformed no additional info (video) and physics only for all models, with mean accuracy gains ranging from 2.7% to 37%.

For **VideoLLaMA3-7B**, the Combined(taxonomy and physics) condition significantly exceeded all other prompting conditions, while physics only and taxonomy only did not differ significantly. For **InternVideo2.5-Chat-8B**, Combined(taxonomy and physics) outperformed no additional info (video) and physics only, and taxonomy only also surpassed no additional info (video); however, physics only and Combined(taxonomy and physics) did not differ significantly. For **InternVL3-2B**, both taxonomy only and Combined(taxonomy and physics) yielded significant improvements over no additional info (video) and physics only, while differences between no additional info (video) and physics only were not statistically significant.

These post-hoc findings provide finer-grained evidence that combining physical parameters with event taxonomy maximally benefits physical reasoning, though the extent of this benefit varies across architectures.

C. Model Success by Question Template and Condition

To enable fine-grained analysis of model performance across question types, we group automatically generated questions into semantically distinct templates using regular-expression pattern matching. The final set of question types is as follows:

- **Contact:** Binary detection of object collisions or contacts.
- **Object Count:** Number of distinct physical entities in the scene.
- **Stop Count:** How many objects came to rest during the video.
- **Rolling Count:** Number of objects displaying rolling motion.
- **Collision Participants:** Number of objects involved in collisions.
- **Top Collider(s):** Which object(s) participated in the most collisions.
- **Kinetic Energy Loss (System):** Percent of energy lost during object-object collisions.
- **Kinetic Energy Loss (Object):** Energy change for individual objects post-collision.
- **Collision Duration:** Time two specific objects remained in contact.
- **Stationary Duration:** Total time an object remained still.
- **Stationary Onset:** Timestamp at which an object first became stationary.
- **Maximum Friction Object:** Object with the highest coefficient of friction.

TABLE V

OVERALL MULTIPLE-CHOICE ACCURACY (%) ACROSS MODELS AND PROMPTING CONDITIONS. EACH MODEL ANSWERS MULTIPLE-CHOICE REASONING QUESTIONS UNDER FOUR PROMPTING CONDITIONS: VIDEO-ONLY, WITH PHYSICAL METADATA (MASS, FRICTION COEFFICIENT, ETC.), WITH TAXONOMY LABELS (E.G. “FRICTION STOP”, “ROLLING”), OR WITH BOTH.

Model	No Additional Info (Video)	Physics Only	Taxonomy Only	Combined (Taxonomy + Physics)
InternVideo2.5-8B	41%	43%	38%	44%
InternVL3-2B	25%	24%	30%	28%
Video-LLaMA3-7B	48%	56%	55%	61%

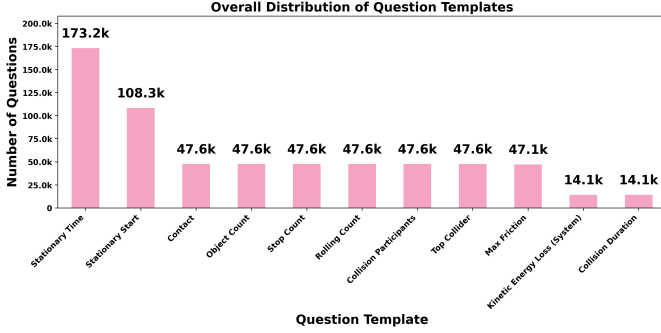


Fig. 3. Overall distribution of question templates in PHLOP (summed over all prompting conditions). Values above the bars are shown in thousands.

Before analysing how each model performs on different question types, it is important to understand how many questions fall into each template. Figure 3 shows the distribution of PHLOP’s automatically generated questions across the 12 semantic templates. Two categories dominate: more than 173 k questions relate to *Stationary Time*, and about 108 k to *Stationary Start*. In contrast, categories such as *Collision Duration* and *Kinetic Energy Loss (System)* each have only above 14 k questions. This imbalance will influence the aggregate accuracy reported later, so it is essential to keep it in mind when interpreting the results.

Each QA pair in PHLOP is tagged with one of these templates, enabling analysis of model strengths and weaknesses. Figure 4 shows the accuracy of each model across question templates and prompting conditions. Each subplot corresponds to a different model: on the left (InternVL3-2B), in the center (InternVideo2.5-Chat-8B), and on the right (Video-LLaMA3-7B). The x-axis indicates question templates described in previous section, while grouped bars compare the four prompting conditions. This layout enables direct comparison of model strengths and sensitivity to prompt structure.

a) Observations:

- **Structured context boosts performance:** All models perform best under the Combined condition, confirming the utility of physics and taxonomy metadata in physical reasoning tasks.
- **Video-LLaMA3-7B dominates in most categories:** Particularly strong on duration-based and attribute-based reasoning (e.g., *Collision Duration*, *Max Friction*).
- **InternVL3-2B prefers taxonomy:** Shows higher performance on discrete or symbolic queries when only

semantic labels are provided, suggesting better alignment with label-based prompting.

- **Collision reasoning remains hard:** Across all models, *Collision Participants* yields low accuracy—even with full metadata—highlighting a limitation in temporal integration or object tracking.

V. LIMITATIONS AND FUTURE WORK

PHLOP currently supports rigid-body dynamics with detailed rule-based annotations. Future extensions should explore deformable bodies, fluid interactions, and differentiable physics engines, significantly broadening the benchmark’s scope and realism (Tung et al. 2023). While PHLOP emphasizes object states and parameterized properties, broader benchmarks such as PhysicsRW (Wang et al. 2024), and Phys-Reason (Liu & et al. 2024) point toward evaluating reasoning under novelty, open-world generalization, and multi-step causal chains. Integrating PHLOP with robotic manipulation tasks and real-world scene generalization could provide further validation of physical reasoning capabilities. Finally, incorporating open-ended language generation and learned classifiers for automatic semantic annotation could also enhance interpretability and generalization, laying the foundations for more robust, physically grounded AI systems. Additionally, future research should evaluate the performance of trained video-language models (Wang et al. 2025, Zhu et al. 2025, Zhang et al. 2025), as current model evaluation is limited to zero-shot VLMs. This evaluation should also examine how physics-aware inductive biases, structured taxonomies, and simulation-grounded data can bridge the gap between perceptual recognition and physically grounded understanding.

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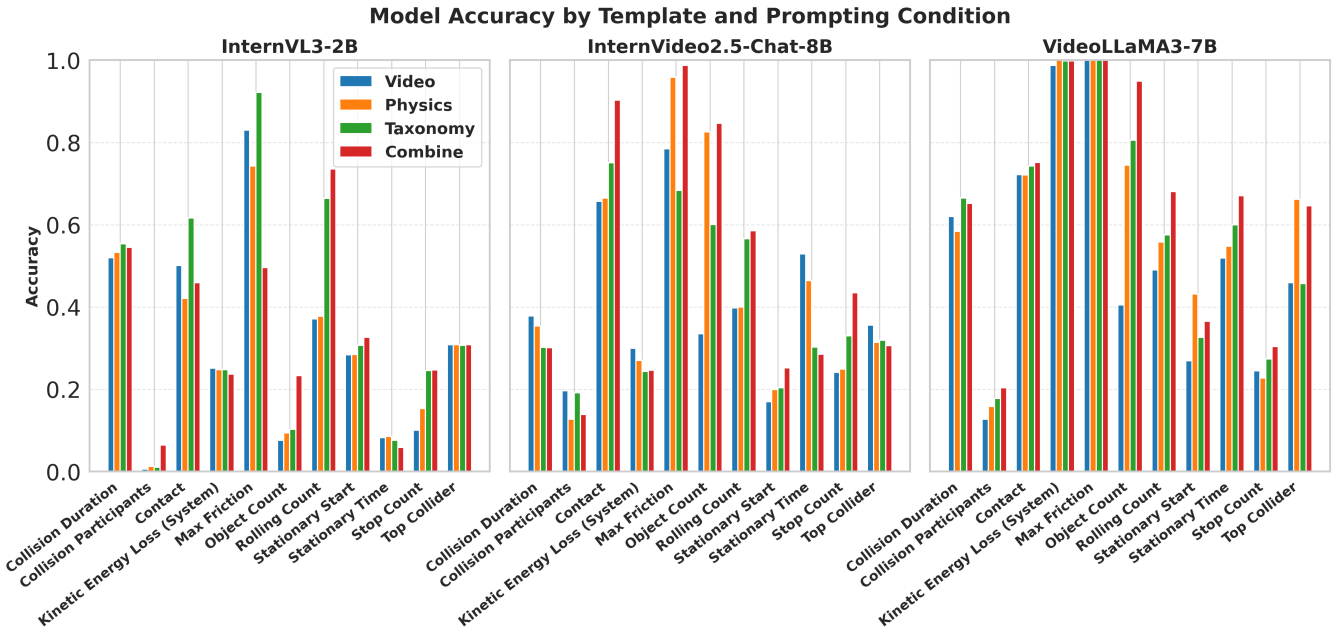


Fig. 4. **Accuracy by question template and prompting condition.** Bars show performance across four input setups for each model. Combining taxonomy and physical metadata consistently improves performance. Collision-related and energy-based tasks remain challenging.

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